

Article

Robust Adaptive Sensorless Control for PMLSM Based on Improved Sliding Mode Observer and Extended State Observer

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Abstract

Nowadays, sensorless control of permanent magnet synchronous linear motors (PMLSM) is widely utilized in industrial applications due to its inherent cost and spatial advantages. However, existing sensorless control methods for PMLMs face insufficient observation accuracy of states and disturbances and poor variable-speed trajectory tracking. To address these issues, this paper proposes a sensorless control method combining multi-observer coordinated perception and robust adaptive control. Firstly, a sliding mode observer based on an improved saturation switching function is designed, which suppresses current noise with a low-pass filter to achieve unbiased estimation of back electromotive force (EMF). Secondly, an extended state observer with back-EMF as input is constructed to synchronously observe disturbances such as the mover speed, position, and thrust ripple of linear machine. Then, a robust adaptive controller is designed to compensate for system uncertainty via an adaptive law, forming closed-loop control with SVPWM. Compared with the traditional methods, the proposed multi-observer coordinated perception scheme can significantly enhance the observation accuracy of the mover speed, position, and lumped disturbances, and the robust adaptive controller can effectively improve the variable-speed trajectory, tracking performance under system uncertainties. Finally, the simulation results have confirmed the effectiveness of the proposed method in accurately observing and tracking speed and position, providing a feasible solution for high-precision sensorless control of PMLSM.

Keywords: permanent magnet linear synchronous motor (PMLSM); sensorless control; sliding mode observer; robust adaptive control



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1. Introduction

PMLSMs are extensively employed in industrial applications, owing to their advantageous characteristics, including rapid dynamic response, high positioning accuracy, and substantial thrust density [1,2]. The acquisition of the position and speed information is the key to the high-precision control of PMLSM [3,4]. Conventional approaches typically rely on mechanical sensors such as photoelectric encoders to directly collect signals, but these methods are often limited by environmental robustness, easily disturbed, and have relatively high purchase and maintenance costs. Consequently, sensorless control technology has emerged as a prominent research direction in linear motor control in recent years [5,6]. This approach offers the distinct advantage of reconstructing motor state information algorithmically, thereby

eliminating the necessity for additional physical sensors. At present, sensorless control methods mainly include the high-frequency signal injection method [7–9], the extended Kalman filter (EKF) [10,11], the model reference adaptive system (MRAS) [12,13], and the sliding mode observer (SMO) [14,15]. Among these, the sliding mode observer has become an important research and application method due to its merits, including low sensitivity to motor model parameter accuracy, robust external disturbance rejection, a relatively straightforward algorithmic structure, and rapid dynamic response.

Traditional SMO suffers from inherent high harmonic components and high-frequency chattering induced by its sign function, which may even cause system instability [16]. To solve this problem, existing studies have introduced low-pass filters and continuous smooth switching functions (e.g., Sigmoid, saturation, hyperbolic tangent functions) to suppress harmonics and noise [17–19], yet conventional SMOs with fixed switching functions in Refs. [14,15] still face an inevitable trade-off between steady-state observation accuracy and chattering suppression. Besides switching function optimization, advanced signal processing technologies, such as adaptive filters and phase-locked loop (PLL), are adopted to improve observation quality [20,21], but the corresponding robust adaptive schemes lack targeted compensation for motor parameter time-variation, leading to poor adaptability to the temperature drift of stator resistance and inductance. In addition, the cascaded SMO-ESO structure has been proposed for disturbance observation and compensation [22,23] and verified in the sensorless control of rotating permanent magnet synchronous motors (PMSMs); however, there is still a lack of mature targeted SMO-ESO sensorless control schemes for permanent magnet synchronous linear motors (PMLSMs). Meanwhile, existing SMO-ESO designs mostly only focus on one-dimensional disturbance compensation, ignoring the coupling effect of system uncertainties, and fail to realize the deep integration of disturbance observation results and control law design.

PMLSM control systems face not only the difficulty in achieving accurate observation of system states but also prominent challenges in disturbance rejection and dynamic response performance. As a result, robust adaptive control strategies have become the preferred solution for motor speed control [24]. By incorporating disturbance rejection terms or designing robust control laws, such strategies reduce the system's sensitivity to thrust force fluctuations, external disturbances, and parameter variations in PMLSMs, thereby enhancing operational stability under uncertain operating conditions [25–28]. Furthermore, the implementation of adaptive laws enables real-time monitoring of system states, facilitating rapid response and precise compensation for disturbances and deviations under complex operating conditions. While this type of method can effectively improve the system's disturbance rejection capability and dynamic response speed, it is plagued by inadequate coordination between the control strategy and the state observer. This limitation readily leads to constrained anti-interference ability, dynamic response lag, and impaired fast synchronization between control outputs and observed states, ultimately degrading the overall dynamic performance of the system.

To address the aforementioned deficiencies in state observation accuracy, disturbance rejection, and dynamic performance of the sensorless PMLSM drive, this work adopts a robust adaptive control strategy based on a cascaded observer structure, which integrates a sliding-mode observer (SMO) and an extended state observer (ESO).

Major contributions and salient features of this work are as follows:

1. A sensorless control scheme combining multi-observer coordinated perception and robust adaptive control is proposed for permanent magnet synchronous linear motors (PMLSM), which addresses the issues of insufficient observation accuracy of states and disturbances, and poor variable-speed trajectory tracking in existing methods.

2. A sliding mode observer (SMO) based on an improved saturation switching function is designed, where a low-pass filter is introduced to suppress current noise, achieving unbiased estimation of back electromotive force (EMF) and significantly alleviating the chattering issue in conventional SMOs.
3. An extended state observer (ESO) with back-EMF as input is constructed to synchronously observe the mover speed, position, and lumped disturbances including thrust ripple, realizing multi-state and disturbance cooperative perception for PMLSM.
4. A robust adaptive controller is designed to compensate for system uncertainties via an adaptive law, which is integrated with SVPWM to form a closed-loop control system, thus significantly improving the variable-speed trajectory tracking performance and anti-interference ability of the PMLSM drive.

The rest of this article is organized as follows. Section 2 presents the mathematical model of PMLSM. The design of the SMO-ESO cascaded observer is introduced in Section 3. Section 4 elaborates on the robust adaptive control strategy. Simulation results and comparative analysis are provided in Section 5. Finally, Section 6 concludes this article.

2. Mathematical Model of PMLSM

The mathematical model of the PMLSM lays a theoretical foundation for the design of its control system. It is derived from the fundamental principles of electromagnetic induction and electromechanical energy conversion. This model explicitly characterizes the dynamic interrelationships among d-q axis voltages, currents, flux linkages, mover velocity, and electromagnetic thrust. Moreover, it incorporates key nonlinear factors inherent to the motor, including parameter variations and end effects, thereby establishing a theoretical basis for the development of high-performance PMLSM control algorithms.

The voltage equation in the dq frame can be expressed as

$$\begin{cases} \dot{i}_d = \frac{1}{L_d} u_d - \frac{R_s}{L_d} i_d + \frac{\pi}{\tau L_d} v \psi_q \\ \dot{i}_q = \frac{1}{L_q} u_q - \frac{R_s}{L_q} i_q + \frac{\pi}{\tau L_q} v \psi_d \end{cases} \quad (1)$$

where u_d and u_q denote the d-axis and q-axis components of the stator voltage, respectively; i_d and i_q represent the d-axis and q-axis components of the stator current; L_d and L_q are the d-axis and q-axis components of the stator inductance; ψ_d and ψ_q correspond to the d-axis and q-axis components of the flux linkage; R_s is the stator resistance; τ stands for the pole pitch; and v indicates the linear velocity of the mover.

The flux linkage equations of the PMLSM are defined as follows:

$$\begin{cases} \psi_d = L_d i_d + \psi_f \\ \psi_q = L_q i_q \end{cases} \quad (2)$$

where ψ_f is the flux linkage of the PMLSM.

The mechanical motion equations of the PMLSM are given by

$$M\dot{v} = F_e - Bv - F_l - F_d \quad (3)$$

$$F_e = P \frac{3\pi}{2\tau} [(L_d - L_q) i_d i_q + \psi_f i_q] \quad (4)$$

where M denotes the mover mass, F_e represents the electromagnetic thrust, F_l is the load force, and F_d is the detent force (cogging force); B represents the viscous friction coefficient, and P is the number of pole pairs.

When the dq-axis inductances are equal ($L_d = L_q = L_s$), the electromagnetic thrust can be simplified as follows:

$$F_e = P \frac{3\pi}{2\tau} \psi_f i_q \quad (5)$$

In this case, the electromagnetic thrust varies linearly with the q-axis current and is independent of the d-axis component.

3. SMO-ESO Velocity and Position Observer

The SMO-ESO combined observer utilizes the advantages of the SMO and ESO to provide accurate observation of the system states and total disturbances, providing theoretical support for the sensorless control technology of PMLSM.

3.1. PMLSM Voltage Equations in the α - β Coordinate Frame

In the α - β stationary coordinate frame, the voltage equations of the PMLSM form the fundamental theoretical basis for the design of the SMO. Utilizing the nonlinear switching mechanism of the SMO, the back-EMF components can be extracted from the stator current dynamics, thus enabling sensorless state observation that is independent of physical sensors.

The voltage equations of the PMLSM in the α - β coordinate frame are given by

$$\begin{cases} \dot{i}_\alpha = \frac{1}{L_s} u_\alpha - \frac{R_s}{L_s} i_\alpha + \frac{1}{L_s} e_\alpha \\ \dot{i}_\beta = \frac{1}{L_s} u_\beta - \frac{R_s}{L_s} i_\beta + \frac{1}{L_s} e_\beta \end{cases} \quad (6)$$

where u_α and u_β represent the α - and β -axis components of the stator voltage, respectively; i_α and i_β denote the α - and β -axis components of the stator current; e_α and e_β are the α and β -axis components of the back-EMF; and L_s is the stator inductance.

The mathematical expressions for the back-EMF components are given by

$$\begin{cases} e_\alpha = -\sin\left(\frac{\pi}{\tau}x\right) * \frac{\pi}{\tau} v \psi_f \\ e_\beta = \cos\left(\frac{\pi}{\tau}x\right) * \frac{\pi}{\tau} v \psi_f \end{cases} \quad (7)$$

where x is the position of the mover.

3.2. Design of the Sliding Mode Observer

The SMO design is based on the mathematical model of PMLSM; it utilizes the inherent robustness of sliding mode variable structure control along with an improved algorithm to facilitate precise and real-time observation of the mover state under sensorless conditions.

A conventional SMO is formulated as

$$\begin{cases} \dot{\hat{i}}_\alpha = \frac{1}{L_s} u_\alpha - \frac{R_s}{L_s} \hat{i}_\alpha + \frac{1}{L_s} \hat{e}_\alpha \\ \dot{\hat{i}}_\beta = \frac{1}{L_s} u_\beta - \frac{R_s}{L_s} \hat{i}_\beta + \frac{1}{L_s} \hat{e}_\beta \end{cases} \quad (8)$$

where $\hat{i}_\alpha$ and $\hat{i}_\beta$ are the observed values of the stator currents i_α and i_β , and $\hat{e}_\alpha$ and $\hat{e}_\beta$ are the observed values of the back-EMF components e_α and e_β .

The observed back-EMF is derived as

$$\begin{cases} \hat{e}_\alpha = K \tanh(\hat{i}_\alpha - i_\alpha) \\ \hat{e}_\beta = K \tanh(\hat{i}_\beta - i_\beta) \end{cases} \quad (9)$$

where $\tanh(ax) = \frac{e^{ax} - e^{-ax}}{e^{ax} + e^{-ax}}$, $K > \max\{e_\alpha \tanh(\hat{i}_\alpha - i_\alpha), e_\beta \tanh(\hat{i}_\beta - i_\beta)\}$; the switching gain K can be adjusted according to the operating speed.

From Equations (6) and (8), the error dynamics can be expressed as

$$\begin{cases} \dot{\hat{i}}_{\alpha} - \dot{i}_{\alpha} = -\frac{R_s}{L_s}(\hat{i}_{\alpha} - i_{\alpha}) + \frac{1}{L_s}(\hat{e}_{\alpha} - e_{\alpha}) \\ \dot{\hat{i}}_{\beta} - \dot{i}_{\beta} = -\frac{R_s}{L_s}(\hat{i}_{\beta} - i_{\beta}) + \frac{1}{L_s}(\hat{e}_{\beta} - e_{\beta}) \end{cases} \quad (10)$$

When the system reaches steady state, $s = \hat{i} - i = 0$, it leads to equality between the observed and actual back-EMF values:

$$\begin{cases} e_{\alpha} = \hat{e}_{\alpha} = K \tanh(\hat{i}_{\alpha} - i_{\alpha}) \\ e_{\beta} = \hat{e}_{\beta} = K \tanh(\hat{i}_{\beta} - i_{\beta}) \end{cases} \quad (11)$$

To mitigate the adverse effects of high-frequency noise, stemming from sliding mode chattering, current sampling interference, and inverter nonlinearities, on the observed back-EMF, and to ensure the accurate subsequent calculation of the mover position and electrical angular velocity, a low-pass filter is employed to further process the observed back-EMF signals. This step effectively suppresses high-frequency noise and generates smoothed back-EMF estimation signals:

$$\begin{cases} \hat{e}_{\alpha} = \frac{\omega_c}{s + \omega_c} K \tanh(\hat{i}_{\alpha} - i_{\alpha}) \\ \hat{e}_{\beta} = \frac{\omega_c}{s + \omega_c} K \tanh(\hat{i}_{\beta} - i_{\beta}) \end{cases} \quad (12)$$

where ω_c is the cutoff frequency of the low-pass filter.

The back-EMF estimation performance of the conventional SMO, the tanh-function-based SMO, and the proposed improved SMO are compared under variable-speed conditions.

The back-EMF waveforms of the PMLSM under a two-step speed profile (accelerating to 0.6 m/s during 0–0.05 s, keeping constant until 0.15 s, then accelerating to 1.2 m/s during 0.15–0.2 s and remaining stable until 0.4 s) are illustrated in Figure 1. It can be clearly observed that the EKF in Figure 1a achieves superior estimation performance, yielding smooth, standard sinusoidal back-EMF with stable amplitude and a well-maintained 90° phase difference throughout the entire operating range. The conventional SMO in Figure 1b suffers from severe chattering and significant square-wave distortion, resulting in poor waveform quality and disrupted phase characteristics. Although the proposed SMO in Figure 1c achieves improved smoothness compared with the conventional SMO, noticeable waveform distortion still appears under the high speed of 1.2 m/s. Three schemes are compared, namely, the SMO with an improved saturation switching function, the SMO with a sign function, and the traditional SMO with a hyperbolic tangent function. The proposed saturation-function-based SMO exhibits the best performance, with fast convergence, smooth sinusoidal waveforms, accurate phase alignment, and stable amplitudes without obvious distortion or performance degradation, which fully demonstrates its higher estimation accuracy and better dynamic tracking performance than the other two SMO methods.

Compared with the traditional sign function and piecewise linear saturation function, the improved saturation function offers core advantages: it suppresses chattering more thoroughly, eliminates discontinuities at transition points, and enhances observation smoothness and steady-state accuracy, while retaining the strong robustness of sliding mode and balancing dynamic response with parameter robustness.

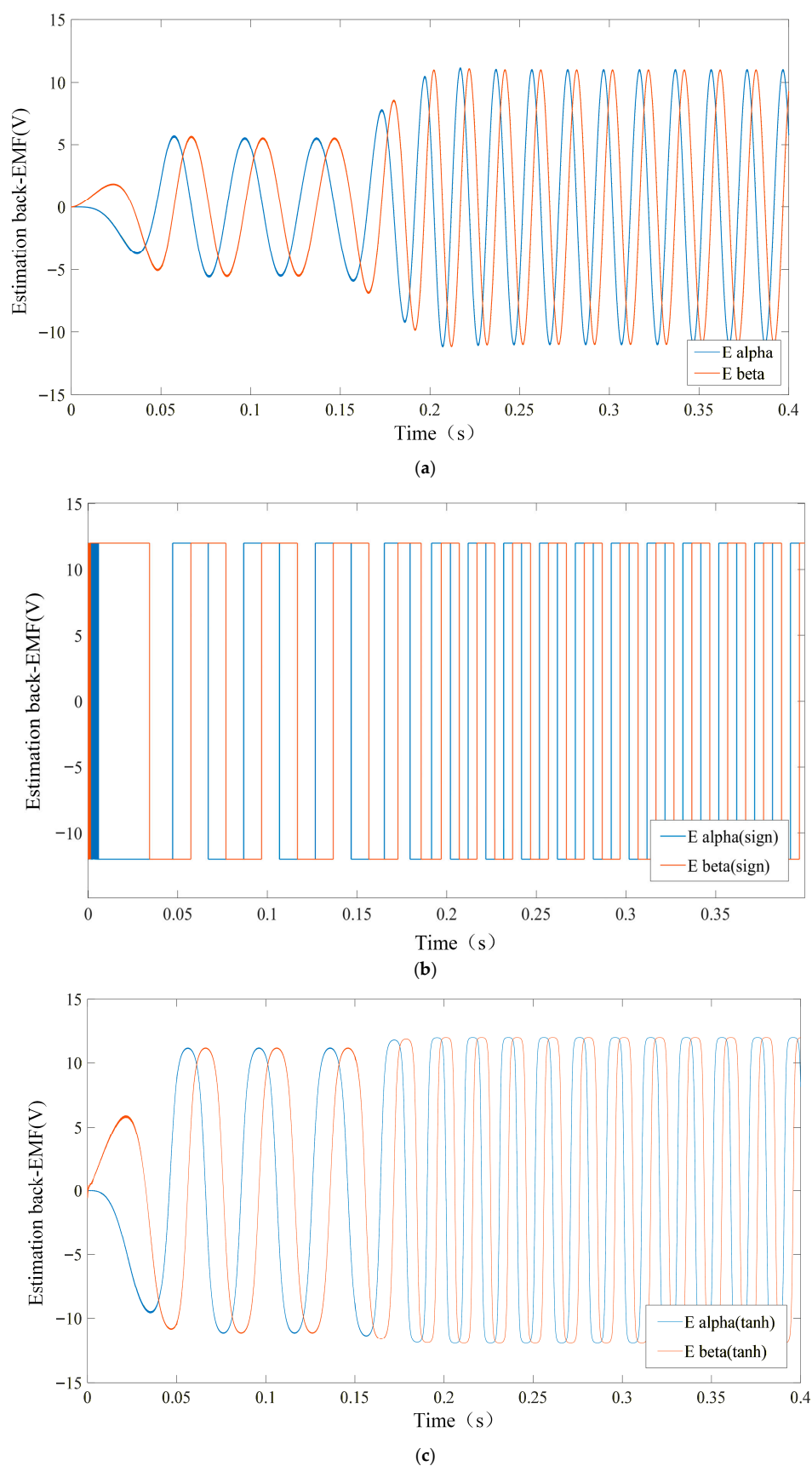


Figure 1. Back-EMF signals estimated by three different SMO. (a) Back-EMF observed by the proposed improved SMO; (b) back-EMF observed by the sign function-based SMO; (c) back-EMF observed by the tanh function-based SMO.

3.3. Parameter Uncertainties and Disturbances

The problems of parameter time-varying, external disturbance, and system coupling faced by PMLSMs under actual working conditions are mainly attributed to the temperature drift characteristics of resistance and inductance, load mutation, and thrust fluctuation of end effect. These uncertainties will directly reduce the dynamic and steady state performance of the system, which requires that the control strategy must have the ability of real-time estimation and dynamic compensation.

By separating the uncertainties and disturbances, the system can be described as

$$\dot{v} = \chi i_q^* + d + g \quad (13)$$

$$\begin{cases} d = \chi(i_q - i_q^*) - \eta v - \gamma(F_l + F_d) \\ g = \Delta\chi i_q - \Delta\eta v - \Delta\gamma(F_l + F_d) \end{cases} \quad (14)$$

where d represents the total disturbance, g denotes the uncertainty, and $\chi = \frac{3\pi P}{2\tau M}\psi_f$, $\eta = \frac{PB}{M}$, $\gamma = \frac{P}{M}$, $\Delta\chi$, $\Delta\eta$, $\Delta\gamma$ gamma are uncertain parameters.

3.4. Design of the Extended State Observer (ESO)

In practice, F_e , F_l , and F_d are difficult to obtain in real time, thereby making it impossible to estimate the motor states via the motor motion equation. Accordingly, this study develops an ESO for the real-time estimation of the aforementioned disturbance and the synchronous observation of the mover position and velocity.

As illustrated in Figure 2, the α - β axis back-EMF signals are estimated using the SMO from the α - β axis stator currents and voltages of the PMLSM. The raw estimated signals are first processed by a low-pass filter, and the filtered back-EMF signals are then fed into the coordinate transformation module to generate the error signal ΔE . Subsequently, this error signal is input into the ESO to realize real-time disturbance observation, as well as simultaneous estimation of the mover position and velocity of the PMLSM. To facilitate the engineering implementation of the proposed observer, a generic second-order extended state observer is designed in this study.

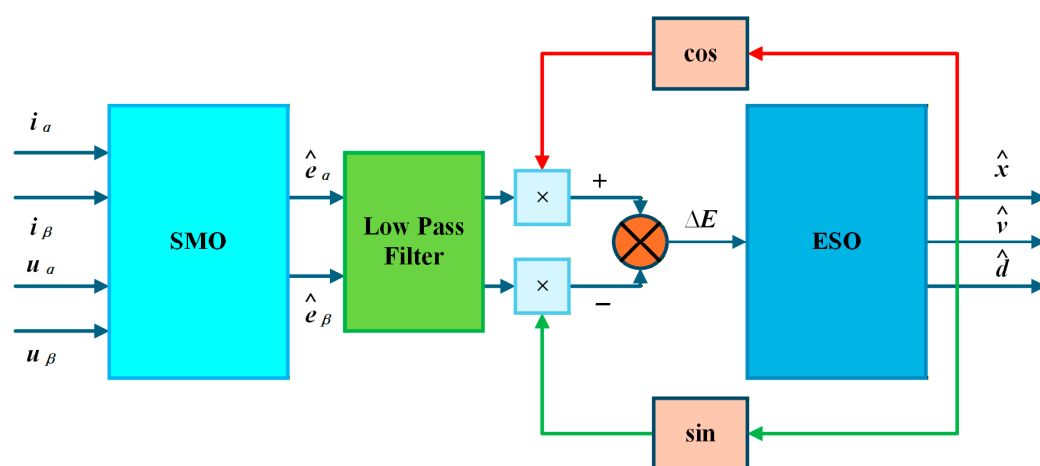


Figure 2. SMO-ESO cascaded structure.

The ESO design may refer to [29], a general second-order ESO can be expressed as

$$\begin{cases} \dot{x}_1(t) = x_2(t) \\ \dot{x}_2(t) = f(x_1, x_2, t) + bu(t) \end{cases} \quad (15)$$

where b is a constant greater than zero, $u(t)$ is the control input, and $f(x_1, x_2, t)$ represents a bounded unknown nonlinear perturbation.

When the observer stabilizes, the position estimation error approaches zero, satisfying $\sin(\frac{\pi}{\tau}\hat{x} - \frac{\pi}{\tau}x) \approx \frac{\pi}{\tau}\hat{x} - \frac{\pi}{\tau}x$. To better process the back-EMF signal, an intermediate variable is defined as follows:

$$\begin{aligned}\Delta E &= -\hat{e}_\alpha \cos \frac{\pi}{\tau}\hat{x} - \hat{e}_\beta \sin \frac{\pi}{\tau}\hat{x} \\ &= k \sin \frac{\pi}{\tau}x \cos \frac{\pi}{\tau}\hat{x} - k \cos \frac{\pi}{\tau}x \sin \frac{\pi}{\tau}\hat{x} \\ &= k \sin(\frac{\pi}{\tau}\hat{x} - \frac{\pi}{\tau}x) \\ &\approx k_1(\hat{x} - x)\end{aligned}\quad (16)$$

Based on Equation (16), the position estimation error can be expressed as follows:

$$e_1(t) = x - \hat{x} = z_1(t) - x_1(t) = -\frac{\Delta E}{k_1}\quad (17)$$

From Equations (13) and (15), by defining $d = x_3 = f(x_1, x_2, t)$, the ESO can be designed as

$$\begin{cases} e_1(t) = \frac{\pi}{\tau}x - \frac{\pi}{\tau}\hat{x} = -\frac{\Delta E}{k_1} \\ \dot{\hat{x}} = \hat{v}(t) - \beta_1 e_1(t) \\ \dot{\hat{v}}(t) = \hat{d}(t) + \hat{g}(t) + \chi i_q^*(t) - \beta_2 e_1(t) \\ \dot{\hat{d}}(t) = -\beta_3 e_1(t) \end{cases}\quad (18)$$

where $\beta_3 = 3\omega$, $\beta_2 = 3\omega^2$, $\beta_1 = \omega^3$, with ω being the observer bandwidth [30]. When $k_1 = \pi k / \tau$, k is the gain parameter. The gain k significantly influences the amplitude and convergence characteristics of the observation error; conversely, a smaller k increases the error amplitude and slows convergence, making it unsuitable for high-precision or high-speed applications. Therefore, the selection of k should be tailored to practical conditions.

4. Robust Adaptive Speed Controller Design

The robust adaptive control strategy combines the technical advantages of adaptive control and robust control and has the features of high synergism with the observer, enabling a trade-off between the anti-disturbance ability and dynamic response performance of the system. Therefore, it provides a high-precision, robust closed-loop control solution for the sensorless control of PMSLMs.

4.1. Design of the Speed Controller

To balance dynamic response and anti-interference ability while reducing electromagnetic thrust fluctuations, the proposed controller comprehensively accounts for velocity error, total disturbances, and system uncertainties in its design.

The speed error is defined as

$$e_v(t) = v^* - \hat{v}\quad (19)$$

where v^* represents the reference speed.

From Equation (13), the dynamic error of speed tracking can be expressed as

$$\dot{e}_v = v^* - \chi i_q^* - \hat{d} - \hat{g}\quad (20)$$

Considering the error characteristics of PMLSM sensorless control, the speed observation error and its integral are incorporated into the sliding surface. By appropriately

constructing the sliding surface parameters, the system error can converge rapidly. The error sliding surface is designed as

$$s = e_v(t) - e_v(0) + c \int_0^t e_v(t) dt, c > 0 \quad (21)$$

To estimate system uncertainties and satisfy stability requirements, the adaptive law is designed as

$$\dot{\hat{g}} = -\lambda s \quad (22)$$

By incorporating the pre-designed sliding surface, the estimated disturbances, and system uncertainties, this paper presents the design of the speed controller as follows:

$$i_q^* = \frac{1}{\chi} \left[\dot{v}^* - \hat{g} - \hat{d} + ce_v + \varepsilon \operatorname{sign}(s) + ks \right] \quad (23)$$

where $k > 0$, and ε is a positive constant. Conventional sliding mode control achieves deterministic robustness via a sign function and fixed sliding surface, but its high-frequency switching term easily causes chattering. In contrast, the proposed method adopts an adaptive law and robust terms to the online estimate and compensates system uncertainties without relying on a fixed sliding surface. With a continuous robust-adaptive structure and no high-frequency switching actions, it inherently suppresses chattering while maintaining strong robustness. Furthermore, its adaptive mechanism improves the adaptability to time-varying uncertainties and external disturbances, leading to smoother control performance and higher steady-state accuracy.

4.2. Stability Analysis of the Controller

To verify the stability and robustness of the proposed robust adaptive controller, an appropriate Lyapunov function based on Lyapunov stability theory is constructed.

The Lyapunov function is selected as

$$V = \frac{1}{2}s^2 + \frac{1}{2\lambda}\hat{g}^2 \quad (24)$$

The uncertainty estimation error is defined as $\tilde{g} = g - \hat{g}$, and the time derivative of the estimation error is $\dot{\tilde{g}} = -\dot{\hat{g}}$. Accordingly, the derivative of V is given by

$$\begin{aligned} \dot{V} &= s\dot{s} + \frac{1}{\lambda}\tilde{g}\dot{\tilde{g}} \\ &= s(\dot{v}^* - \chi i_q^* - \hat{d} - \hat{g} + ce_v + \varepsilon \operatorname{sign}(s) + ks) - s(\varepsilon \operatorname{sign}(s) + ks) - \tilde{g}s + \frac{1}{\lambda}\tilde{g}\dot{\tilde{g}} \\ &= s(\varepsilon \operatorname{sign}(s) + ks) \end{aligned} \quad (25)$$

When the parameters are selected to satisfy $\varepsilon > |\dot{\tilde{v}}| + |\dot{\tilde{d}}|$, $\tilde{v} = v - \hat{v}$, and $\tilde{d} = d - \hat{d}$, then $\dot{V} \leq 0$ is established. According to the Lyapunov stability theorem, the robust adaptive speed controller designed in this paper is asymptotically stable.

5. Results and Analysis

In this section, comprehensive simulation verifications are conducted to quantitatively evaluate the performance of the proposed control strategy from four critical indicators, namely observer estimation accuracy, dynamic response speed, steady-state control precision, and anti-interference ability, which demonstrates its comprehensive superiority. Figure 3 depicts the schematic diagram of the PMLSM control system.

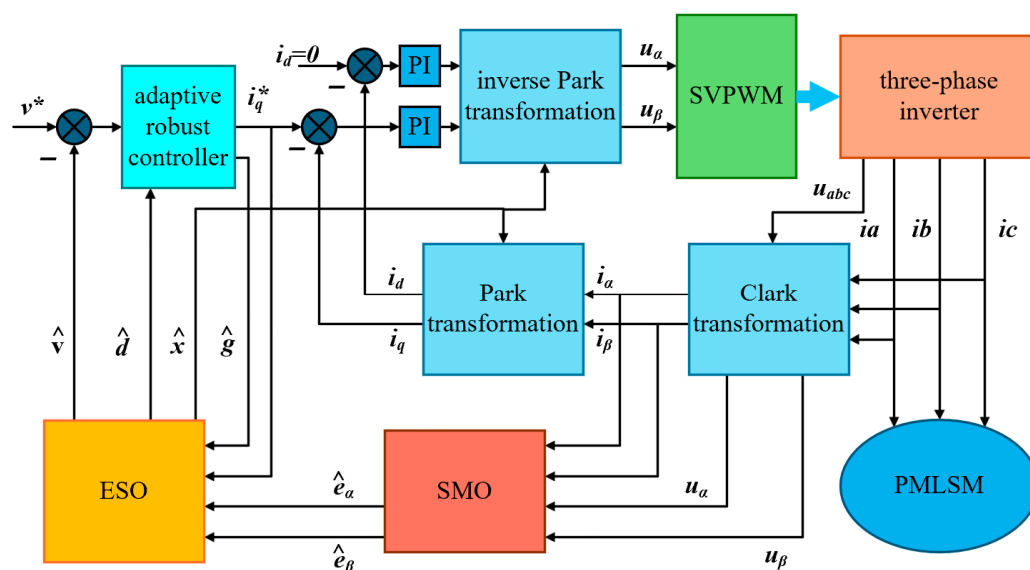


Figure 3. Diagram of the PMLSM control system.

Notably, the integrated configuration of the SMO-ESO and the proposed robust adaptive control algorithm constitutes the distinctive innovation of this control system, with the developed control algorithm implemented in the speed loop to govern the dynamic response and anti-interference ability. The parameters of the PMLSM utilized for the simulations are detailed in Table 1.

Table 1. Motor parameters.

Symbol	Denotation	Value	Unit
L_d	d-axis inductance	0.00123	H
L_q	q-axis inductance	0.00123	H
R_s	stator resistance	6.5	ohm
ψ_f	permanent magnet flux	0.0385	Wb
τ	pole pitch	0.012	m
M	mover mass	1.51	kg
B	viscous friction coefficient	0.001	N s/m
P	number of pole pairs	1	Pair

5.1. Analysis of Parameter Variation Effects on Control Accuracy

The analysis diagrams of the parameter variation effects are added to investigate the impacts of $\pm 20\%$ variations in R_s , L_s , and B on control accuracy under steady-state conditions at the speeds of 0.6 m/s and 1.2 m/s, with a variation step of 10%.

To evaluate the robustness of the proposed control strategy against parameter variations, we conducted tests under $\pm 10\%$ and $\pm 20\%$ perturbations of the viscous friction coefficient B , stator inductance L_s , and stator resistance R_s , respectively. The speed response curves under these conditions are shown in Figure 4a–c. As shown in Figure 4a, an increase in B reduces the overshoot during the speed response but slightly prolongs the rise time. Conversely, a decrease in B increases the overshoot but accelerates the response, while the steady-state speed remains stable within a narrow range, indicating the system's good robustness to B variations. As shown in Figure 4b, an increase in L_s leads to more pronounced oscillations in the speed response, a larger overshoot, and a longer settling time, whereas a decrease in L_s results in a faster response with reduced oscillations, demonstrating that the system is more sensitive to increases in L_s . As shown in Figure 4c, an increase in R_s slightly prolongs the rise time and increases the overshoot, while a decrease

in R_s accelerates the response and reduces the overshoot; the overall impact of R_s variations on the system's steady-state performance is minimal, confirming the strategy's robustness against resistance changes.

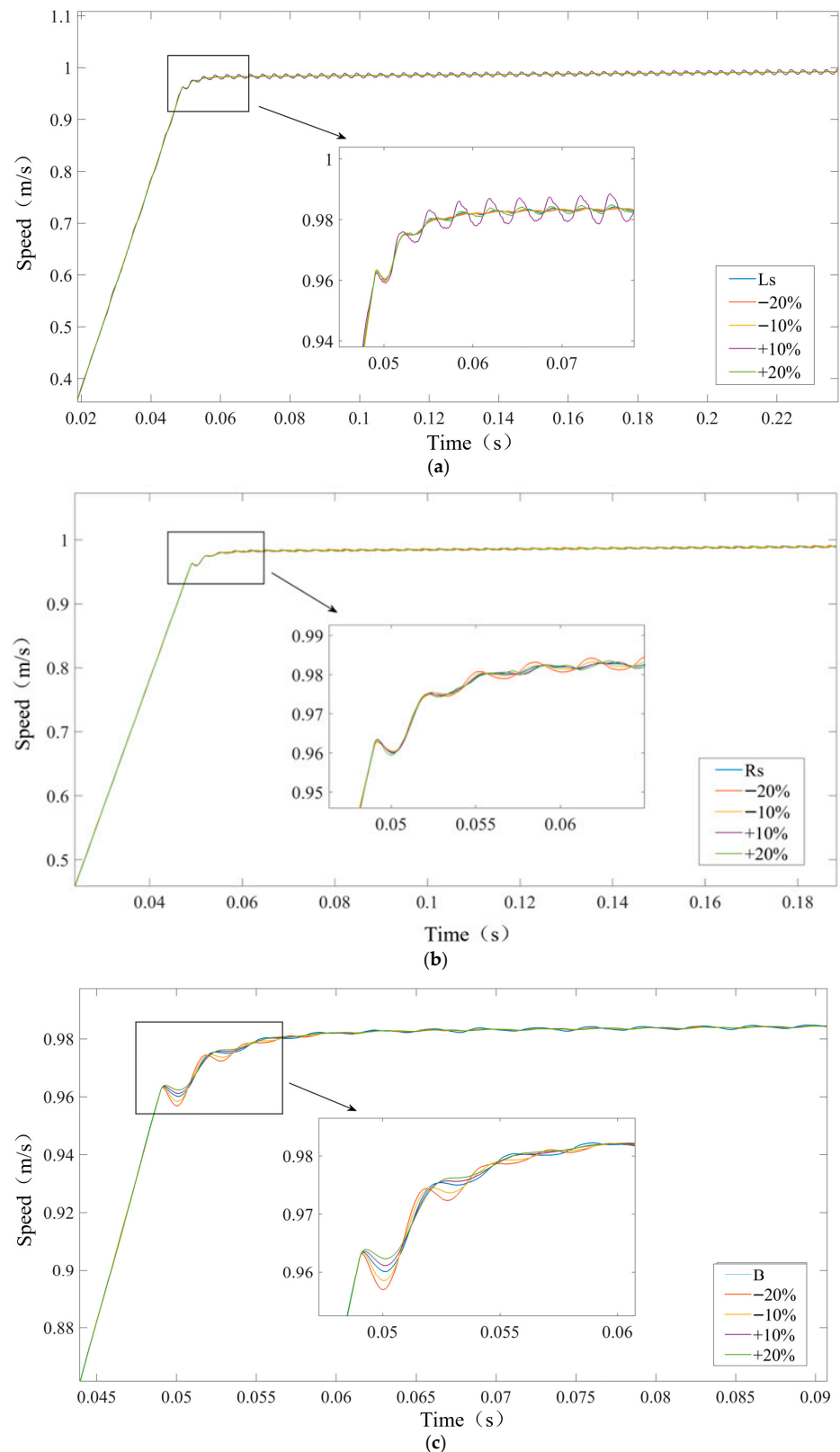


Figure 4. Parameter variation effects on control accuracy. (a) Effect of L_s parameter perturbation on the dynamic response of the system; (b) effect of R_s parameter perturbation on the dynamic response of the system; (c) effect of B parameter perturbation on the dynamic response of the system.

Table 2 is obtained through a series of simulation tests. Among all parameter perturbations as Table 2, an increase in L_s has the most significant impact on the system, leading to increased overshoot and prolonged settling time, which indicates that the system is more sensitive to inductance increments. In contrast, variations in R_s and B have a milder effect on the system, with the steady-state speed error remaining below 0.0023 m/s across all test cases, which fully demonstrates the excellent robustness of the proposed control strategy against parameter variations.

Table 2. Parameter variation effects on control accuracy.

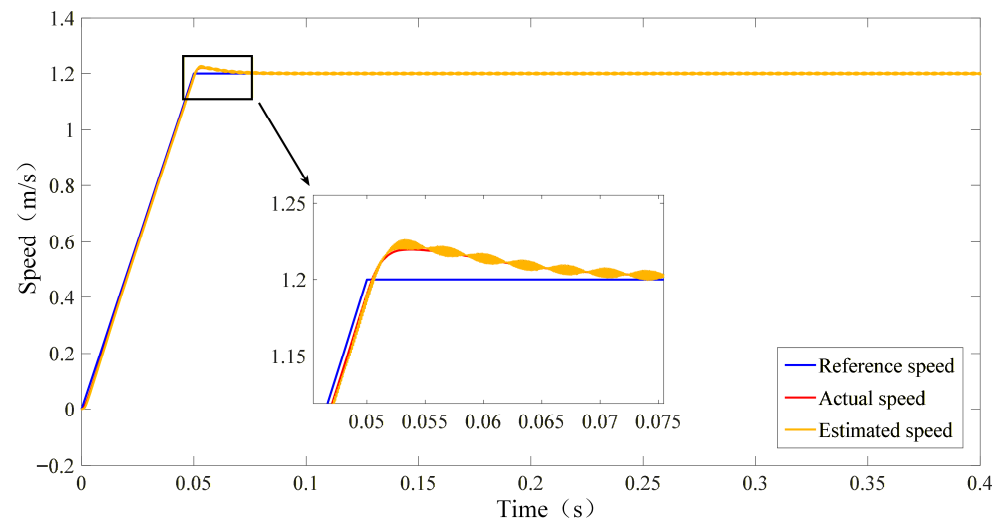
Parameter	Perturbation	Rise Time (s)	Overshoot OS (%)	Settling Time (s)
Nominal	0	0.010	<0.1	0.010
B	−20%	0.009	2.2	0.009
B	−10%	0.0095	2.0	0.0095
B	+10%	0.011	1.7	0.011
B	+20%	0.012	1.5	0.012
L_s	−20%	0.009	1.8	0.009
L_s	−10%	0.010	2.3	0.010
L_s	+10%	0.011	2.5	0.012
L_s	+20%	0.012	2.7	0.013
R_s	−20%	0.009	1.5	0.009
R_s	−10%	0.0095	1.7	0.0095
R_s	+10%	0.011	2.0	0.011
R_s	+20%	0.012	2.3	0.012

5.2. Comparison Between Sensed and Sensorless PI Control

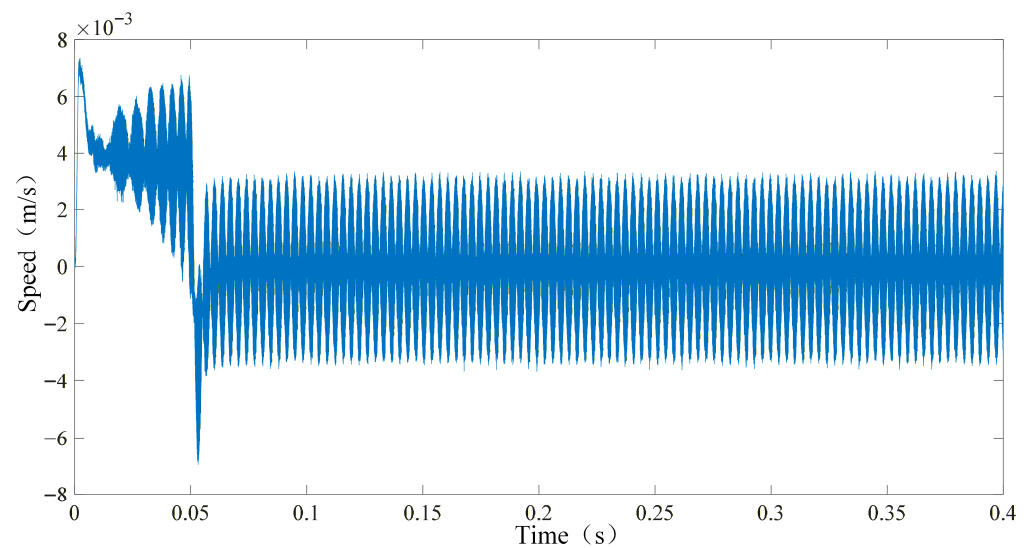
In order to validate the high estimation accuracy of the designed observers, a comparative analysis between the sensed PI control and sensorless PI control strategies is conducted.

In Simulation 1, the reference speed was set to Speed Curve 1 as shown in Figure 5a, and a conventional PI controller [31] was employed under both sensed and sensorless conditions. As depicted in Figure 5a, during the dynamic response phase, the estimated speed from the sensorless PI control tightly tracks the actual speed, with a maximum dynamic deviation of merely 0.0041 m/s; in the steady state, the two curves nearly overlap, achieving a coincidence degree exceeding 99.9%. The speed estimation error curve in Figure 5b further reveals that the error remains in narrow range, with peak-to-peak fluctuations limited to below 0.0062 m/s; after the system reaches dynamic stability at 0.2 s, the error amplitude is further reduced to less than 0.0037 m/s, maintaining an average steady-state error of only 0.0019 m/s. These data demonstrate that the speed estimation signal generated by the sensorless PI control scheme is highly accurate and reliable for practical engineering applications.

Figure 6 illustrates the speed comparison between the sensorless and sensed PI control. The speed curves of the sensorless PI control and sensed PI control are highly consistent; the steady-state speed deviation of the sensorless control is 0.0023 m/s, which is comparable to the 0.0018 m/s of the sensed control. These results confirm that the proposed sensorless PI control scheme exhibits high estimation accuracy, with its speed tracking performance on par with that of the sensed counterpart.



(a)



(b)

Figure 5. Sensorless PI control under the condition of time-varying speed reference command. (a) Comparison between the estimated speed and the actual speed; (b) speed estimation error under PI sensorless control.

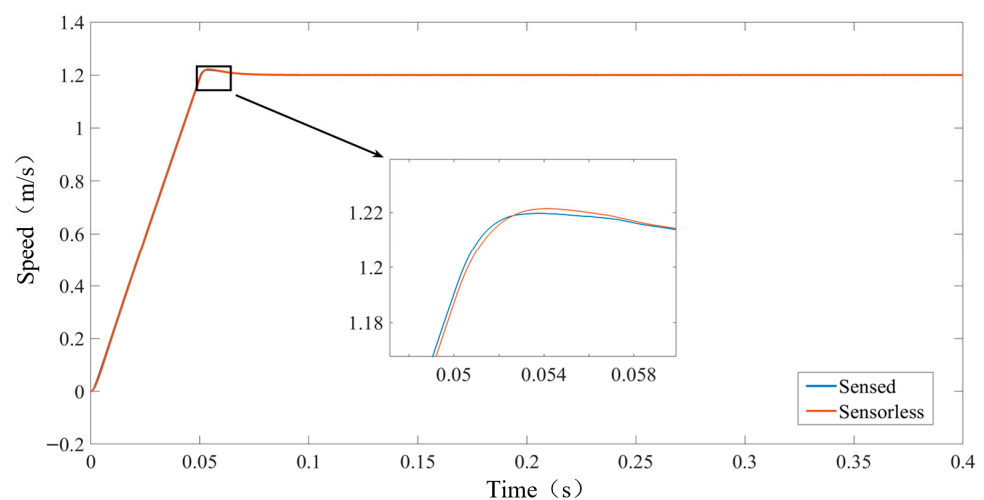


Figure 6. Speed comparison between PI sensorless and PI sensed control.

5.3. Sensorless Robust Adaptive Control

To verify the effective coordination between the control module and the observation module, the sensorless robust adaptive control system is operated under two typical operating conditions, namely time-varying speed and external disturbance.

In Simulation 2, the sensorless robust adaptive control strategy was implemented, with the reference speed initially set to Speed Curve 2, as shown in Figure 7a.

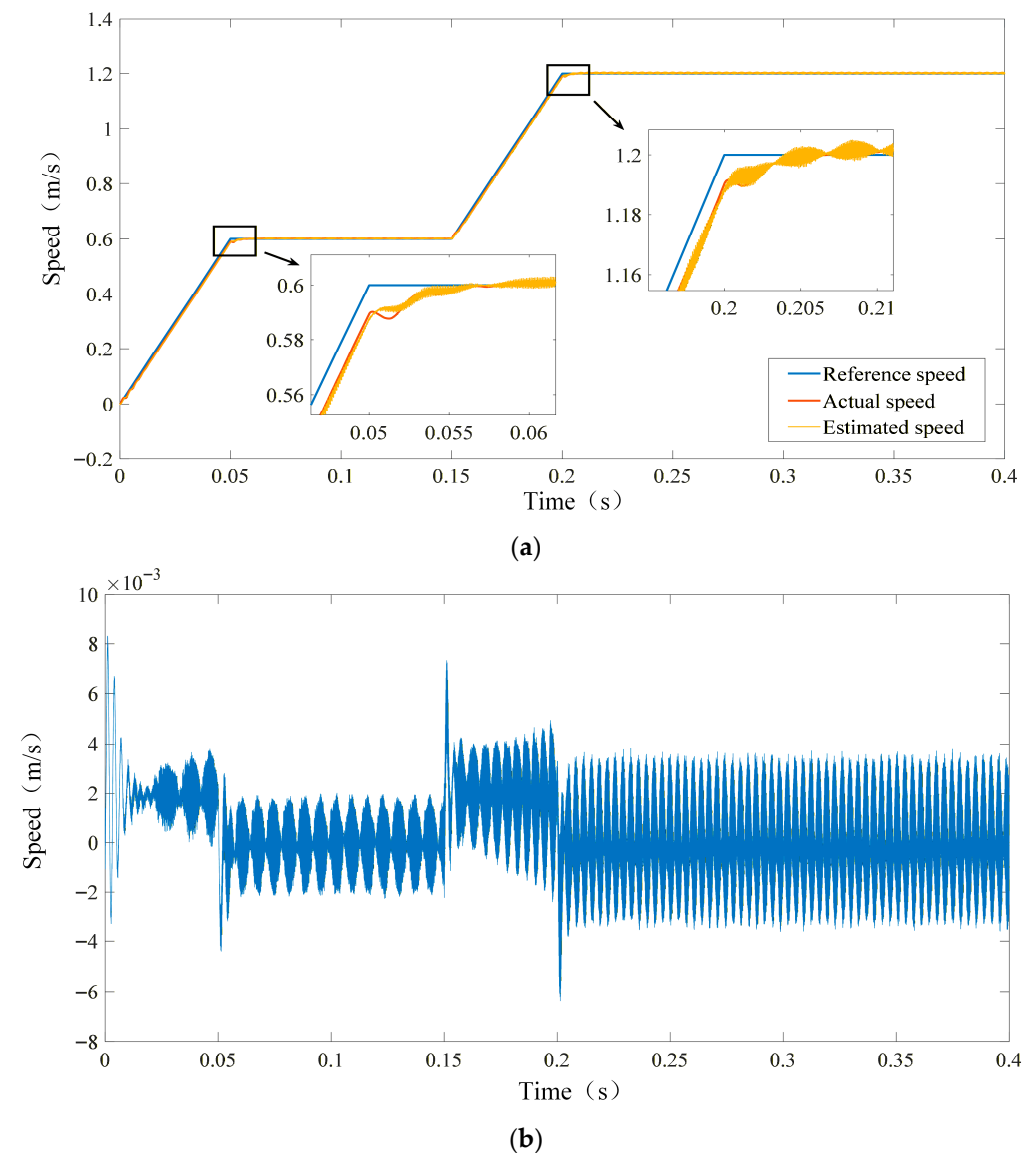


Figure 7. Sensorless robust adaptive control under the operating conditions of time-varying speed reference command. (a) Comparison between the estimated speed and the actual speed; (b) speed estimation error under sensorless robust adaptive control.

As shown in Figure 7a, the control scheme exhibits exceptional speed tracking performance, during the 0–0.05 s and 0.15–0.2 s. The overshoot is nearly zero less than 0.1%, and the actual and estimated speeds rise smoothly without any oscillations. In the steady state, the estimated, actual, and reference speeds achieve an almost complete overlap, with a maximum steady-state deviation of 0.0021 m/s. As presented in Figure 7b, the speed tracking error throughout the simulation remains below 0.008 m/s, with an average error of 0.0034 m/s, a value practically negligible in engineering applications. These data confirm that the control scheme achieves a fast dynamic response while maintaining high tracking accuracy and operational smoothness.

Figure 8 illustrates the speed tracking performance of the proposed control strategy during the speed zero-crossing process under a triangular speed profile. It can be observed that the actual speed and the estimated speed closely follow the reference speed throughout the entire operation, especially during the zero-crossing phase around 0.1 s magnified in the inset. Even at the speed reversal point, the deviation between the estimated speed and the actual speed remains extremely small, demonstrating the excellent zero-crossing tracking capability of the proposed method. Figure 8 presents the corresponding speed estimation error curve over the full simulation time domain. The estimation error is primarily concentrated within a narrow range, with transient peaks occurring only during the speed reversal and acceleration/deceleration stages. In the steady-state phase after 0.15 s, the error exhibits stable and small-amplitude oscillations, which fully verifies the high estimation accuracy and strong stability of the proposed observer.

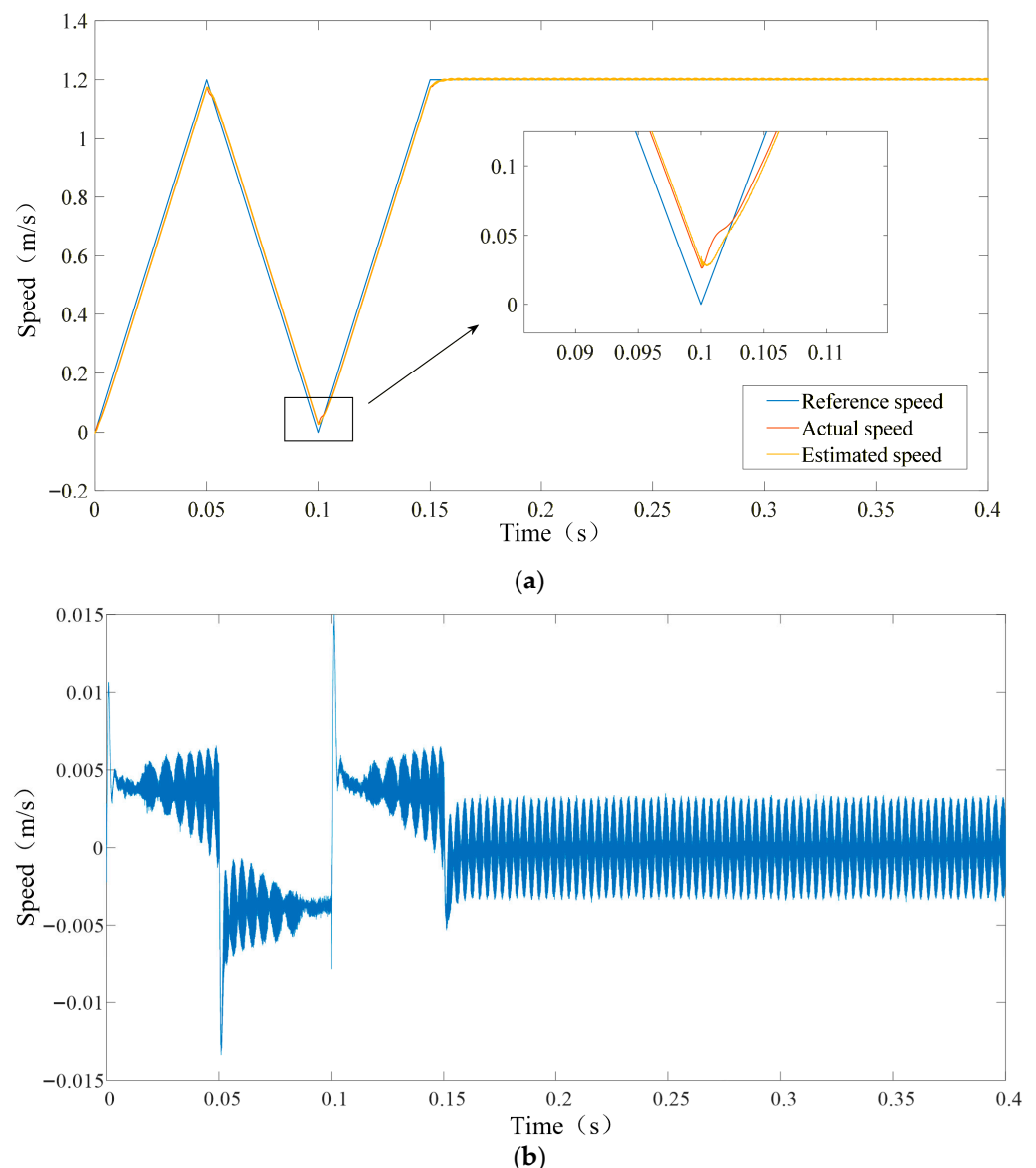


Figure 8. Sensorless robust adaptive control under the operating conditions of zero-crossing speed curve reference command. (a) Dynamic response of zero-crossing speed curve tracking; (b) observation error of the zero-crossing speed tracking curve.

Figure 9a presents the speed tracking performance of the proposed control strategy under a 5 N load disturbance applied at 0.2 s and removed at 0.3 s during the zero-crossing

speed process. It can be observed that the actual speed and estimated speed can still closely follow the reference speed even under sudden load variations, with a maximum transient speed deviation of only 0.03 m/s at the disturbance application moment and 0.025 m/s at the disturbance removal moment, as shown in the magnified insets. This demonstrates the strong anti-interference ability of the proposed method. Figure 9b shows the corresponding speed estimation error curve. The estimation error is maintained within a narrow range of ± 0.03 m/s, with only transient peaks at the disturbance application and removal points and quickly returns to a stable small-amplitude oscillation below ± 0.003 m/s in the steady state, which fully verifies the high accuracy and robustness of the proposed observer under load disturbances.

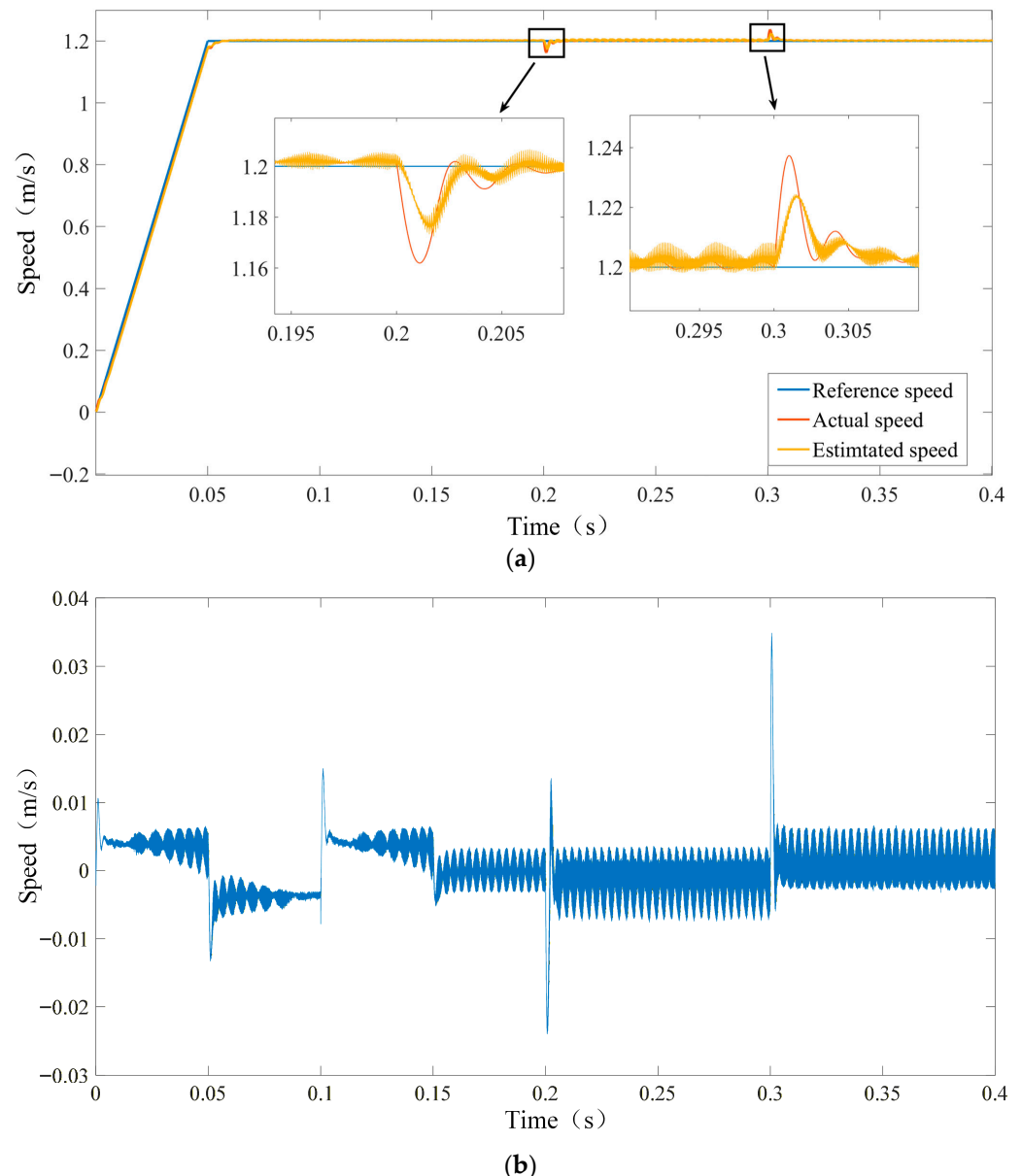


Figure 9. Sensorless robust adaptive control under the operating conditions of zero-crossing speed curve reference command under the conditions of external disturbances. (a) Speed tracking performance under load disturbance during the zero-crossing process; (b) corresponding speed estimation error.

Figure 10 illustrates the speed tracking performance of the proposed sensorless control strategy under a speed switching profile, where the reference speed changes from 1.2 m/s to -1.2 m/s at 0.05 s. It can be observed that both the actual speed and the estimated speed

can closely follow the reference speed during the entire speed switching process, with only a small transient deviation at the speed reversal point. The maximum transient speed deviation is less than 0.03 m/s, and the system quickly recovers to steady-state operation with a stable speed error below ± 0.003 m/s, which fully demonstrates the excellent tracking accuracy and dynamic response capability of the proposed method under rapid speed switching conditions.

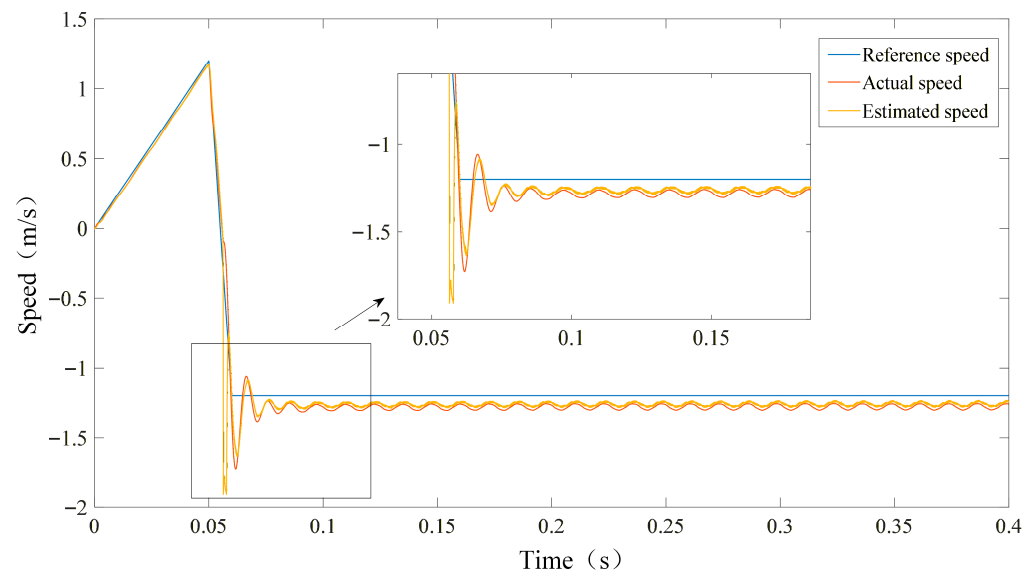


Figure 10. Speed tracking performance of the proposed method under a speed switching profile.

Figure 11 evaluates the speed tracking performance of the proposed control strategy under specific discrete and time-delay constraints. As shown in Figure 11a, under the discrete operation condition with a fixed sampling period of 0.001 s, the actual speed and estimated speed closely track the reference speed throughout the entire process. The maximum transient deviation is limited to 0.02 m/s during the acceleration phases, and the steady-state tracking error is strictly confined within ± 0.003 m/s, which validates the excellent adaptability of the proposed method to high-frequency discrete sampling. As depicted in Figure 11b, when the system operates under the combined conditions of 0.001 s discrete sampling and 0.01 s time delay, the tracking performance exhibits slight degradation but remains satisfactory. A maximum transient deviation of 0.03 m/s occurs at the speed step transition points, 0.05 s and 0.2 s, yet the system rapidly regains stability, with the steady-state error maintained within ± 0.004 m/s. These results fully demonstrate the strong robustness of the proposed control strategy against the combined effects of discrete sampling and time delay.

Figure 12 illustrates the speed tracking performance of the proposed sensorless control strategy under Gaussian white noise. It can be observed that the actual speed and estimated speed still closely follow the reference speed, with only minor fluctuations in the steady-state phase. The maximum speed deviation is maintained below 0.005 m/s, demonstrating the good noise suppression capability of the proposed method. Figure 13 presents the speed tracking performance under a periodic disturbance. The system exhibits stable tracking behavior, with the actual speed and estimated speed effectively suppressing the periodic fluctuations. The steady-state error remains within ± 0.004 m/s, which verifies the strong robustness of the proposed control strategy against periodic disturbances.

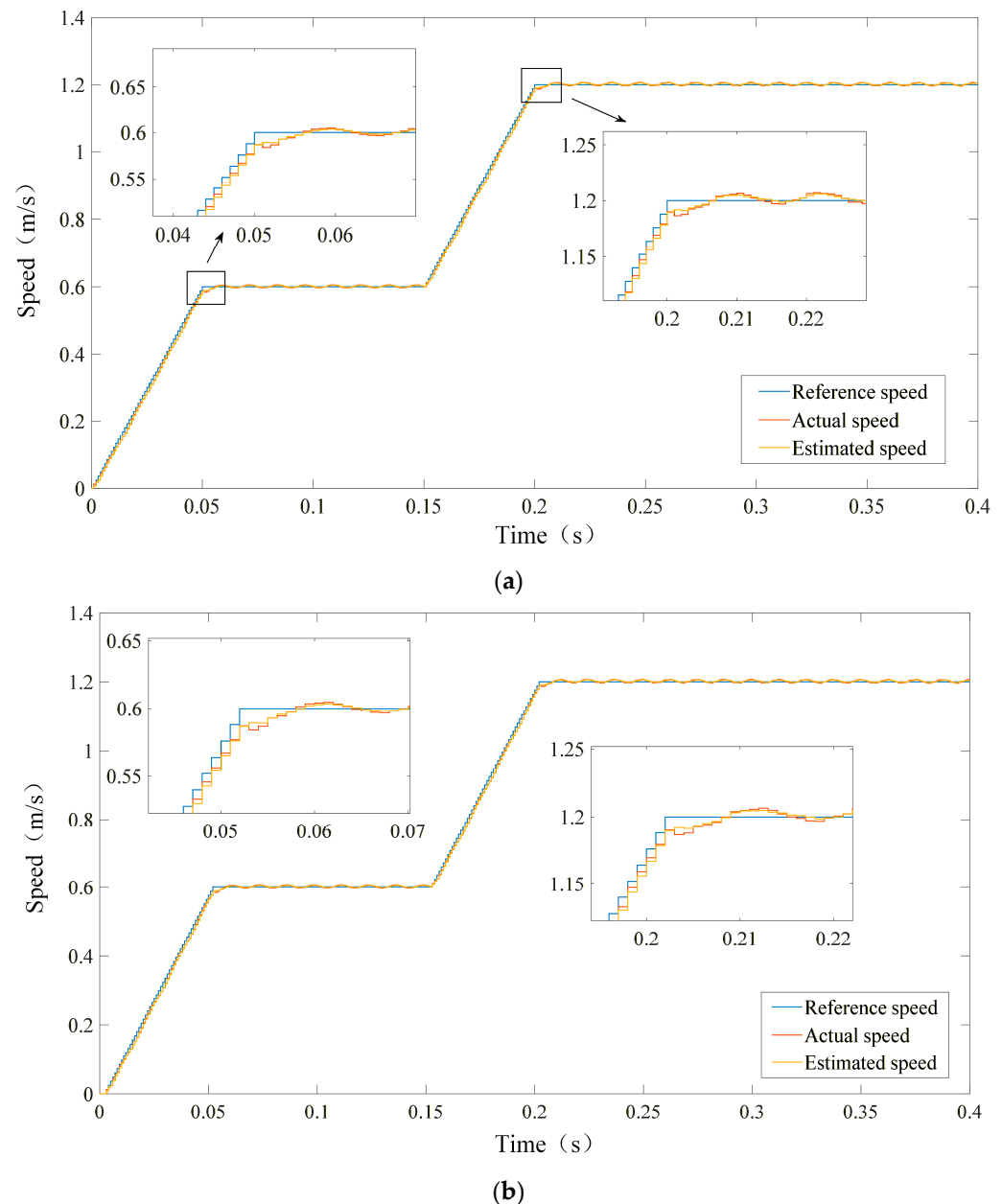


Figure 11. Sensorless robust adaptive control under the conditions of discrete and time-delay operation. (a) Speed tracking performance under discrete operation at sampling period: 0.001 s; (b) speed tracking performance under discrete 0.001 s and time-delay 0.01 s operation conditions.

In a further test under Speed Curve 1, an external disturbance of 5 N was applied at 0.2 s and removed at 0.3 s. Figure 14a highlights the excellent anti-interference ability: upon disturbance application at 0.2 s, the actual speed exhibits a slight fluctuation with a maximum deviation of only 0.017 m/s from the reference speed, and it recovers to within 0.003 m/s of the reference value within approximately 0.0048 s. When the disturbance is removed at 0.3 s, the speed deviation is limited to 0.012 m/s, and full recovery is achieved in 0.0042 s. As shown in Figure 14b, under disturbed operating conditions, the maximum estimation deviation is less than 0.025 m/s, and the estimation deviation remains below 0.01 m/s for most of the remaining time. These results demonstrate that the system can effectively suppress speed fluctuations and achieve rapid recovery from external disturbances, thus exhibiting superior robustness and stability.

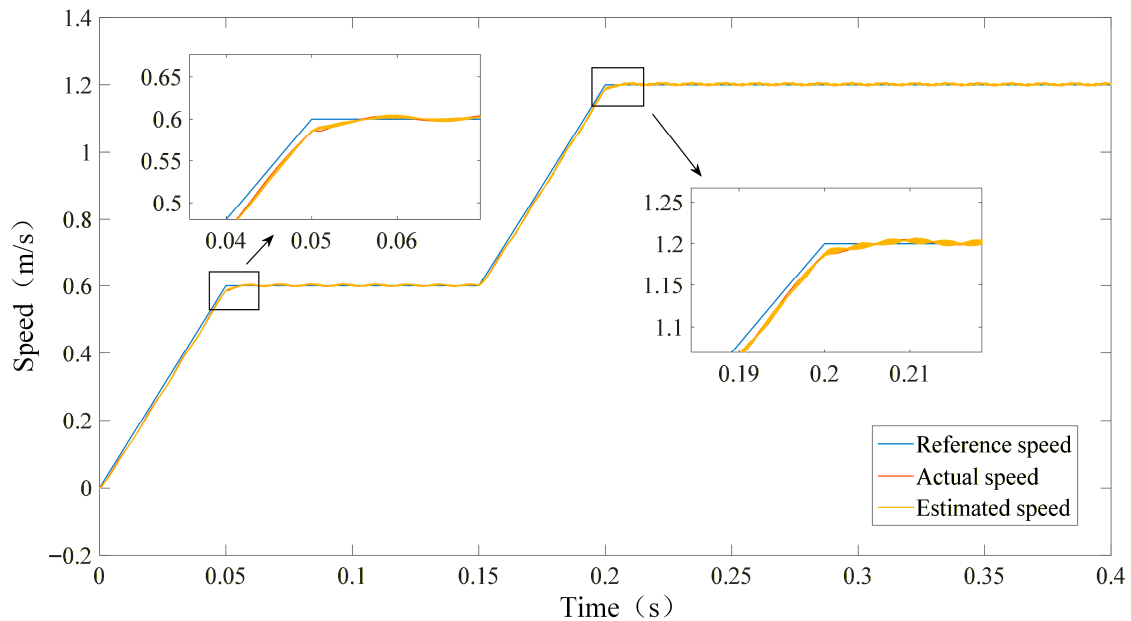


Figure 12. Speed tracking performance under noise.

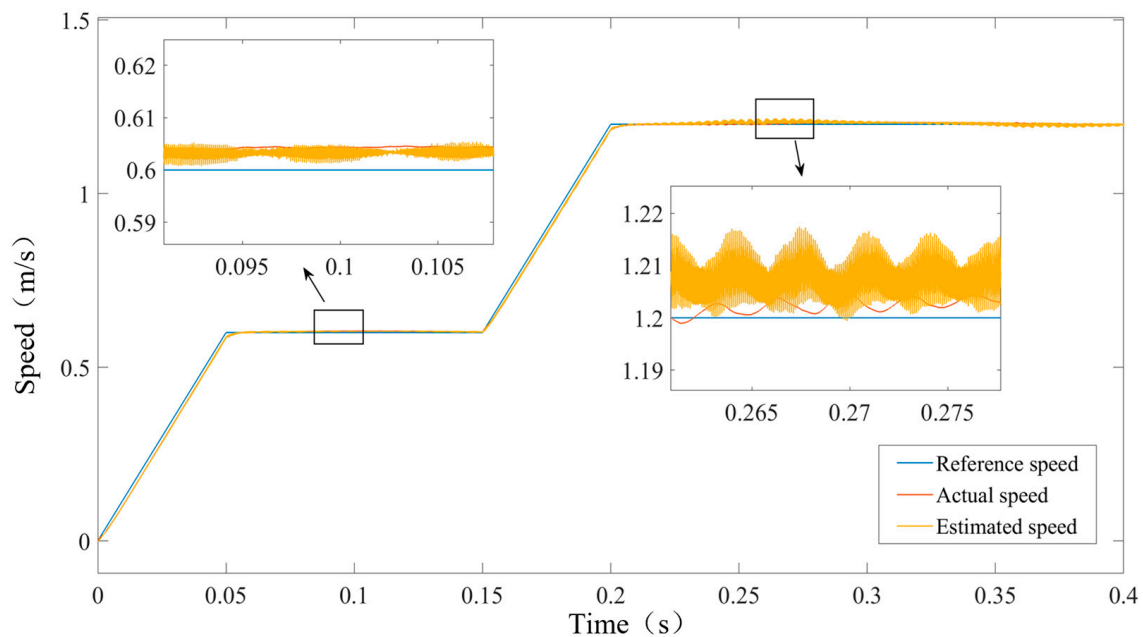


Figure 13. Speed tracking performance under periodic disturbance.

Figure 15 clearly illustrates that, at the instant the 5 N external disturbance is applied at 0.2 s, the estimated total disturbance rises sharply and stabilizes within a value range corresponding to the actual disturbance magnitude. When the disturbance is removed at 0.3 s, the estimated value quickly reverts to its initial stable range. This synchronous response and accurate tracking of the disturbance state confirm the high reliability and precision of the disturbance estimation algorithm. As demonstrated by the comprehensive results of Simulation 2, the SMO-ESO integrated observer exhibits high coordination with the robust adaptive controller, endowing the system with a rapid dynamic response and superior anti-interference ability.

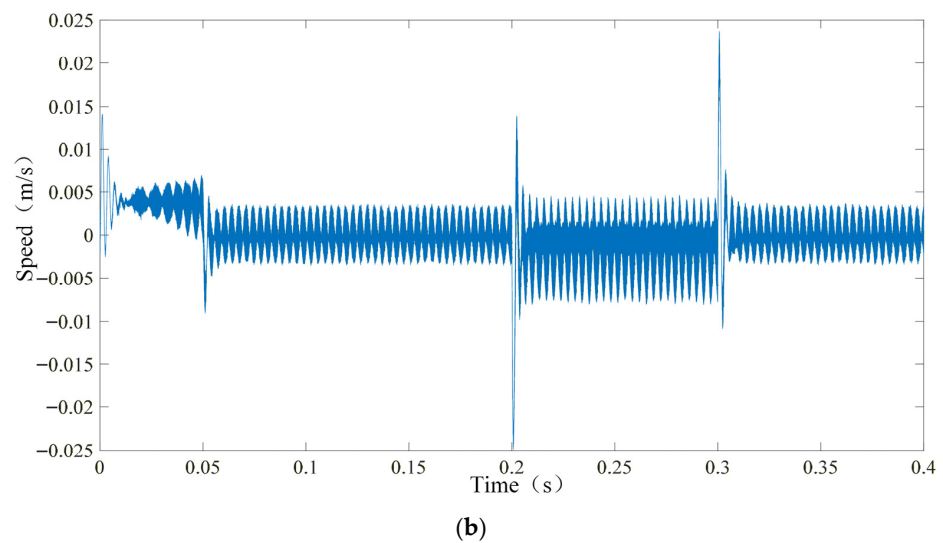
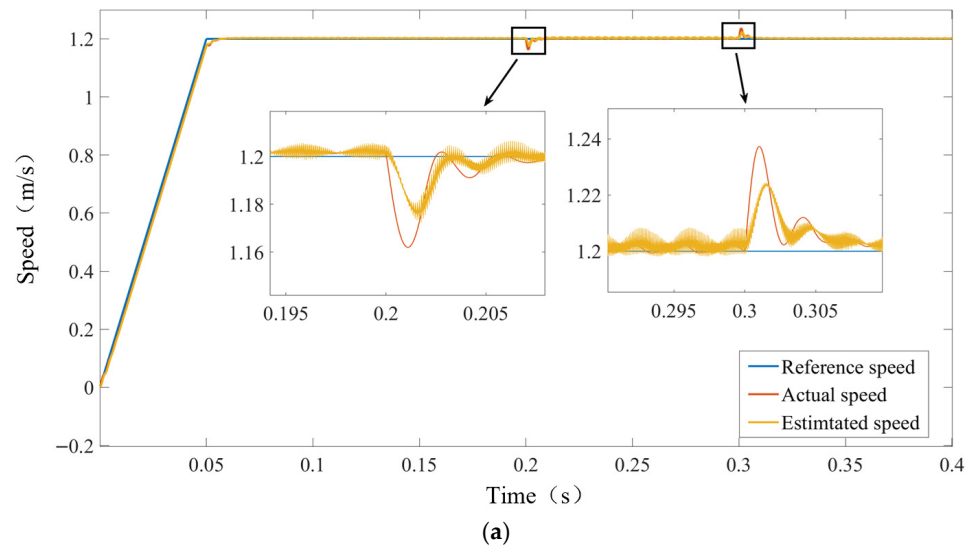


Figure 14. Sensorless robust adaptive control under the conditions of external disturbances. (a) Comparison between the estimated speed and the actual speed; (b) speed estimation error under sensorless robust adaptive control.

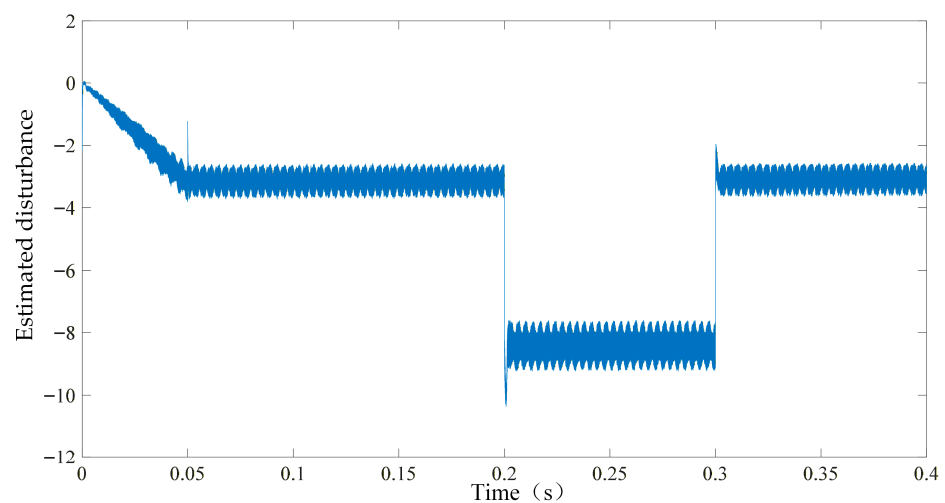


Figure 15. Estimated disturbance under robust adaptive sensorless control.

To quantitatively validate the effectiveness of the proposed control strategy, the core performance evaluation indicators, including disturbance estimation error (DEE), coinci-

dence degree (CD), recovery time (RT), disturbance suppression efficiency (DSE), speed estimation error (SEE), rise time (Tr), overshoot (OS), settling time (Ts), and steady-state speed error (SSE), are defined and summarized in Table 3.

Table 3. Dynamic response performance indicators.

Abbreviation	Full Name	Core Definition & Purpose	Mathematical Definition
DEE	Disturbance Estimation Error	Evaluates ESO's lumped disturbance tracking accuracy; normalized for cross-condition comparability	$DEE(t) = \frac{\widehat{d}(t) - d(t)}{d(t)} \times 100\%$
CD	Coincidence Degree	Quantifies matching degree between estimated and actual speed; reflects observer's global estimation consistency	$CD = \left(1 - \frac{\ \widehat{v}(t) - v(t)\ _2}{\ v(t)\ _2}\right) \times 100\%, t \in [0, T]$
RT	Recovery Time	Evaluates system anti-interference ability; characterizes recovery speed after external disturbance	
DSE	Disturbance Suppression Efficiency	Quantifies anti-interference superiority of the proposed method over baseline controls	$DSE = \left(1 - \frac{\Delta v_{proposed}}{\Delta v_{baseline}}\right) \times 100\%$
SEE	Speed Estimation Error	Evaluates SMO-ESO's real-time speed estimation accuracy for PMLSM mover	$SEE(t) = \hat{v}(t) - v(t)$
Tr	Rise Time	Standard indicator for system step response rapidity	
OS	Overshoot	Evaluates system stability and damping characteristics in step response	$OS = \frac{v_{max} - v^*}{v^*} \times 100\%$
Ts	Settling Time	Comprehensive indicator for system dynamic-to-steady-state transition speed	
SSE	Steady-State Speed Error	Evaluates system steady-state control accuracy	$SSE = \frac{1}{T_{ss}} \int_{T_{ss1}}^{T_{ss2}} v(t) - v(t) dt$

Table 4 summarizes the dynamic response metrics of the proposed method under nine typical operating scenarios. Overall, the method exhibits excellent comprehensive performance: it achieves a fast dynamic response with a settling time of 0.01–0.015 s across all scenarios, maintains high steady-state accuracy with a steady-state error below 0.005 m/s, and has a maximum speed deviation no more than 0.03 m/s. It shows strong anti-interference ability against load, step, noise, and periodic disturbances, with a rapid recovery time of 0.0048–0.006 s when disturbed. Additionally, it demonstrates good adaptability to discrete sampling and time-delay constraints, as well as reliable tracking performance in zero-crossing and speed reversal operations, fully verifying its suitability for high-precision motion control applications.

Table 4. Dynamic response performance indexes under various operating scenarios.

Operating Scenario	Settling Time (s)	Steady-State Error (m/s)	Max Speed Deviation (m/s)	Recovery Time (s)
Startup	0.01	<0.0034	0.02	-
Zero-crossing speed tracking	0.012	<0.003	0.03	0.005
Zero-crossing with load	0.015	<0.0034	0.025	0.0048

Table 4. Cont.

Operating Scenario	Settling Time (s)	Steady-State Error (m/s)	Max Speed Deviation (m/s)	Recovery Time (s)
Discrete	0.011	<0.003	0.02	-
Discrete + delay	0.013	<0.004	0.03	0.006
Speed reversal	0.014	<0.003	0.03	0.0055
Noise	0.01	<0.005	0.005	-
Periodic disturbance	0.012	<0.004	0.02	0.005
Step disturbance	0.015	<0.0034	0.025	0.0048

5.4. Ablation Study via Simulation

The performance of the proposed robust adaptive control strategy was compared with that of PI control and sliding mode control. As shown in Figure 16, during the transition from the acceleration phase to the steady state, the robust adaptive control yields a response curve that closely tracks the target value with minimal oscillations, an overshoot of less than 0.2%; in contrast, conventional PI control exhibits a noticeable overshoot of 3.7%, with steady-state oscillations of ± 0.015 m/s; sliding mode control shows a slightly larger overshoot of 0.8% and peak-to-peak fluctuations of 0.019 m/s. Overall, the robust adaptive control strategy demonstrates a smoother dynamic response, smaller overshoot, higher tracking accuracy, and superior robustness for PMLSM sensorless operation.

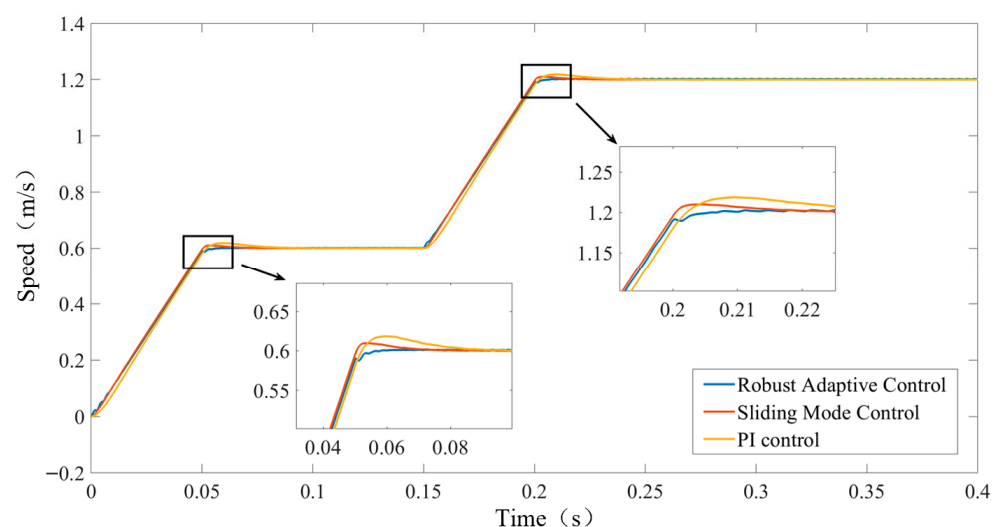


Figure 16. Speed tracking curves with different control methods under the operating conditions of time-varying speed reference command.

Figure 17 presents the speed responses under a 5 N external disturbance of 5 N was applied at 0.2 s and removed at 0.3 s. Upon disturbance application, PI control exhibits significant oscillations with a maximum speed dip of 0.089 m/s and a recovery time of 0.032 s. The sliding mode control shows a larger speed dip of 0.043 m/s and a recovery time of 0.018 s, and the robust adaptive control strategy demonstrates minimal speed fluctuation and the fastest recovery. After disturbance removal, the PI control sustains oscillations for 0.027 s with a maximum deviation of 0.063 m/s. The sliding mode control undergoes a relatively slower recovery with a deviation of 0.028 m/s, and the robust adaptive control strategy only exhibits slight oscillations and reverts to the target speed in 0.0042 s, with a disturbance suppression efficiency improved by over 75% compared to the PI control and over 55% compared to the sliding mode control. These data confirm that the robust

adaptive control strategy outperforms both conventional PI control and traditional sliding mode control in terms of anti-interference ability.

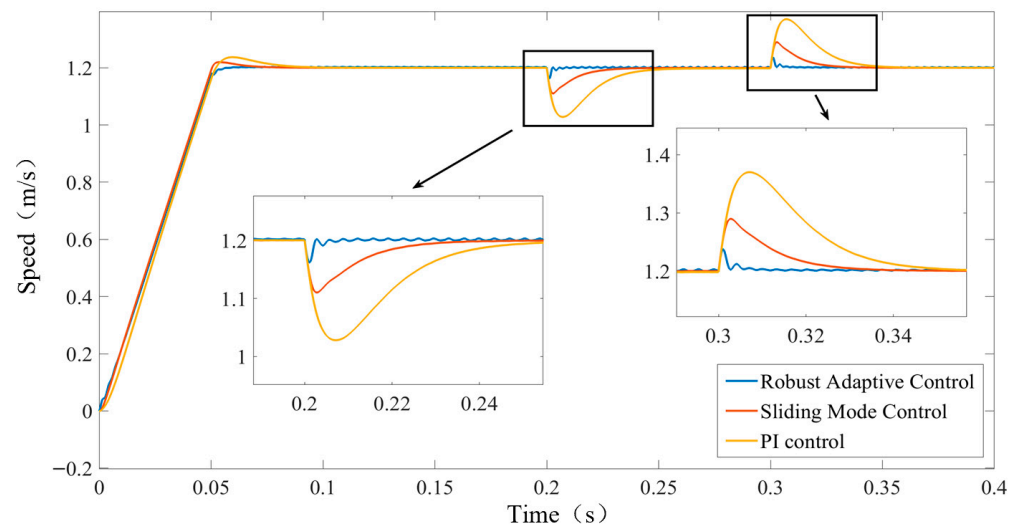


Figure 17. Comparison of speed tracking curves of different control methods under external disturbance operating conditions.

As shown in Figure 18, when the proposed SMO-ESO observer is replaced by an extended Kalman filter (EKF) for speed estimation, the overall speed tracking performance is inferior to that of the SMO-ESO. During the acceleration phase (0–0.05 s), the EKF-estimated speed exhibits noticeable lag and chattering, leading to a larger transient deviation from the reference speed. At the speed step change at 0.2 s, the EKF-based estimation shows a pronounced overshoot and a prolonged recovery time, with the maximum transient deviation reaching 0.08 m/s, which is significantly larger than that achieved by the SMO-ESO. In the steady-state phase (after 0.3 s), the EKF estimation still suffers from persistent small-amplitude oscillations, resulting in a steady-state error of approximately ± 0.005 m/s, which is also inferior to the stable and precise tracking of the SMO-ESO. These results clearly demonstrate that the proposed SMO-ESO observer outperforms the EKF in both transient response and steady-state accuracy, especially under dynamic speed variations.

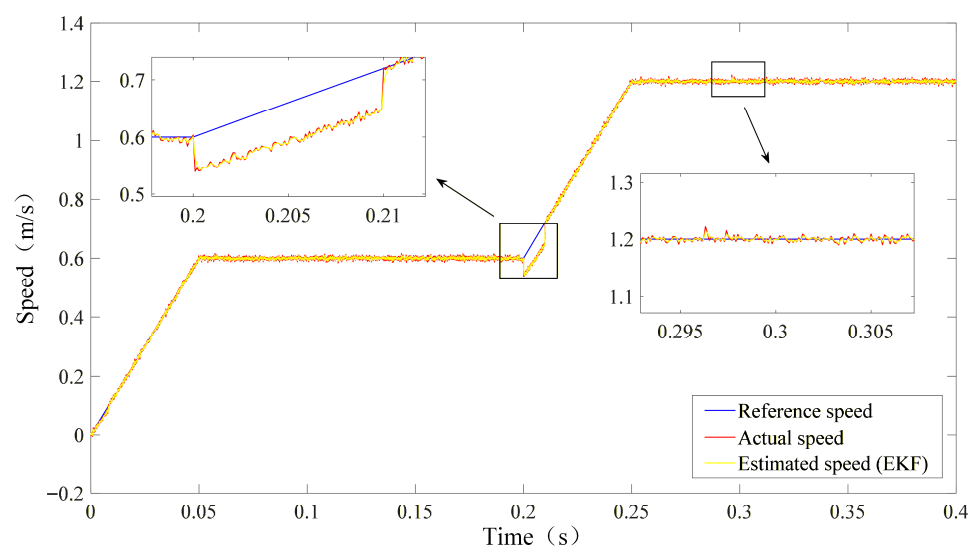


Figure 18. Speed tracking performance of the system when the proposed SMO-ESO observer is replaced by an extended Kalman filter (EKF).

6. Conclusions

To address the challenges of parameter uncertainty, inadequate state observation accuracy, and limited anti-interference ability in PMLSM control systems, this paper proposes a sensorless robust adaptive control method based on SMO-ESO integration. Simulation results validate the correctness and effectiveness of the proposed method. The integrated observation SMO-ESO achieves high-precision estimation of position, speed, and total disturbances. The speed estimation error is below 0.008 m/s, and the disturbance estimation error is less than 1.4%. The proposed robust adaptive controller significantly improves the comprehensive performance: the dynamic response settling time is reduced to 0.01 s, the steady-state speed error is maintained below 0.0034 m/s, and the maximum speed deviation under 5 N external disturbance is only 0.025 m/s with a recovery time of 0.0048 s. Compared to a conventional PI control and traditional sliding mode control, the proposed method reduces overshoot by more than 94% and 78%, respectively, shortens recovery time under disturbance by over 85% and 63%, and enhances anti-interference ability by over 75% and 55%. These results confirm that the proposed sensorless robust adaptive control method effectively mitigates the effects of PMLSM parameter uncertainties and external disturbances, providing a solution for high-precision motion control applications.

However, the proposed method still has certain challenges. It exhibits degraded performance in ultra-low speed tracking scenarios, and noticeable observation fluctuations occur during speed direction switching. Experimental verification will be conducted in future work.

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